Regulatory Capture via Relative Prices

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FSMA, 2016

In 2016 the Food Safety Modernization Act, the largest overhaul of U.S. food-safety law since 1938, came into force. The largest food manufacturers had supported and lobbied for it: a law that raised their own compliance costs.

- The justification came from the 2008–09 Peanut Corporation of America *Salmonella* outbreak. Over seven hundred sickened, nine dead.
- **Interestingly, the act would not have prevented the outbreak.** PCA already ran the monitoring FSMA later made mandatory. It ran *Salmonella* tests, and shipped despite the tests coming back positive, under falsified certificates. Its executives were convicted under statutes that *predate* FSMA.

Why would firms lobby for a rule that raises their own costs?

This is a pattern

- **Amazon** has asked Congress to double the federal minimum wage.
- **Apple** has called on Congress for stricter data-privacy law.
- **Meta** has asked Congress for content-moderation legislation.

How does a firm benefit from a regulation that makes production more expensive?

Intuition

- These rules have *asymmetric incidence* over the firms they bind.
- Every price rises, but the prices of the firms with the least incidence rise *least*.
- They become *relatively* cheaper, demand reallocates toward them, and they grow.

Despite being intuitive, this mechanism has not been studied in the literature. Marginal-cost-raising regulation with asymmetric incidence is studied through the homogeneous-good lens of *raising rivals' costs*; and when a cost-raising regulation is studied in differentiated goods, it affects only fixed costs. In neither setting do relative prices move.

FSMA: who was already compliant

The text that passed largely codifies the practices of the largest manufacturers: hazard analysis, environmental monitoring, hygienic zoning. Systems they had run since the 1990s, because exporting to the EU and Japan required them.

- For those firms, the regulation was cheap: it mostly mandated what they already did. **The firms whose practices became the text are the ones that supported and lobbied for it.**
- For smaller firms, the text meant new personnel and new programs applied on every production run, resulting in a higher marginal cost.

The regulations I study

This paper studies regulations with regressive incidence over marginal costs: incidence that falls with productivity.

- The top firms already run the required systems, use less of the regulated input, or pay above the proposed floor. Meta already runs the industry's largest moderation apparatus; Apple's profits do not rely on targeted ads; Amazon already pays near the proposed floor.
- They increase inequality in the size distribution of firms: large firms grow (gain revenue, profit, and market share); smaller firms shrink.

- Within industries, size across firms has grown steadily more unequal for at least a century (Kwon et al., 2024). The regulations this paper studies could be part of the reason.
- The labor share (Autor et al., 2020), worker inequality (Song et al., 2019), aggregate markup (De Loecker et al., 2020), and aggregate volatility (Gabaix, 2011) depend on the size distribution of firms.

PE: A differentiated industry

Static industry, unit continuum of varieties $\omega \in (0,1)$; homothetic demand with a single price aggregator $P(\mathbf{p})$, gross substitutes (Matsuyama–Ushchev 2017). Firm ω has constant marginal cost ψ_ω and chooses its price:

$$\max_{p_\omega} \ \pi_\omega \ = \ \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y \, s_\omega\left(\frac{p_\omega}{P(\mathbf{p})}\right).$$

The same problem, written in the relative price $z_\omega \equiv p_\omega / P(\mathbf{p})$:

$$\max_{z_\omega} \ \pi_\omega \ = \ \left(1 - \frac{\psi_\omega / P(\mathbf{p})}{z_\omega}\right) Y \, s_\omega(z_\omega)\,.$$

Relative marginal cost ψ_ω / P is the sufficient statistic for the firm's outcomes: in equilibrium it determines its relative price, its market share, its revenue, and its profit.

Who wins

Firms are ranked by ω, so firms become more productive as they approach $\omega = 0$. A **regulation** is a deformation of the marginal-cost profile, $\psi \to \psi^R$. Its *incidence* on firm ω: $1 + \varepsilon_\omega \equiv \psi^R_\omega / \psi_\omega$.

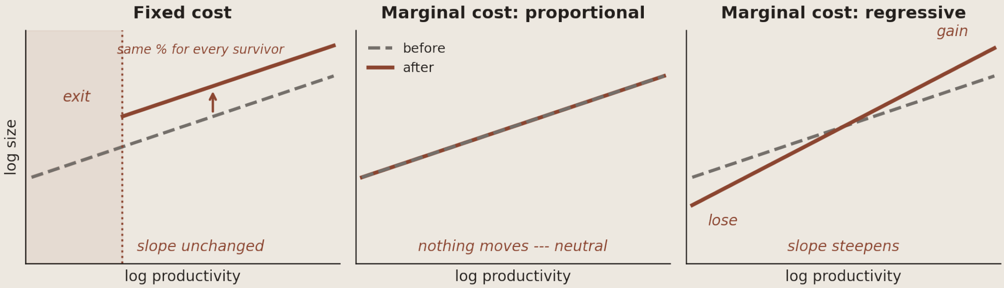
Proposition (Regressive incidence)

Suppose incidence is *regressive* — falling with productivity. Then there exist cutoffs $0 < \bar{\omega} \leq \bar{\omega}' < 1$ such that every firm in $(0, \bar{\omega})$ strictly gains revenue, profit, and market share, and every firm in $(\bar{\omega}', 1)$ strictly loses.

- The firms with the least incidence see their relative marginal cost *fall*: the price environment they compete against rises more than they do, and demand reallocates toward the top.

Observing regressive incidence

Plot log size against log productivity: the slope is the *productivity–size gradient*. The type of incidence in costs deforms the gradient in its own way:



Only regressive incidence steepens the gradient.

CES: Incidence as an elasticity

Specialize to CES,

$$\ln r_\omega \;=\; \underbrace{\ln Y + (\sigma-1)\ln\frac{P}{\mu\, e(r)}}_{\text{common to all firms}} \;+\; \underbrace{(\sigma-1)}_{\text{the gradient}}\;\ln\varphi_\omega, \qquad \psi_\omega = e(r)/\varphi_\omega.$$

A regulation multiplies each firm's cost by $(1+\varepsilon_\omega)$. The gradient moves by $\sigma-1$ times the *elasticity of incidence* t:

$$\Delta\,\text{gradient} \;=\; (\sigma-1)\times t, \quad t \;\equiv\; -\frac{d\ln(1+\varepsilon_\omega)}{d\ln\varphi_\omega}$$

regressive incidence $\Rightarrow \boldsymbol{t > 0}$ progressive $\Rightarrow t < 0$

proportional, selection through fixed costs $\Rightarrow t = 0$

Measuring the elasticity

Productivity is not observed, but baseline revenue is a *model-correct* proxy:

$$\ln r^0_\omega \;=\; a_0 \;+\; (\sigma-1)\ln\varphi_\omega \qquad \Longleftrightarrow \qquad \ln\varphi_\omega \;=\; \frac{\ln r^0_\omega - a_0}{\sigma-1}\,.$$

With a constant elasticity of incidence, the revenue change is linear in baseline size,

$$\Delta\ln r_\omega \;=\; \text{const} \;+\; t\,\ln r^0_\omega,$$

and the coefficient on baseline revenue is the elasticity itself. The nuisance term $(\sigma-1)$ cancels:

$$\beta \;=\; \frac{\operatorname{Cov}\big(\Delta\ln r_\omega,\,\ln r^0_\omega\big)}{\operatorname{Var}\big(\ln r^0_\omega\big)} \;=\; \frac{(\sigma-1)\,t\,\cdot\,(\sigma-1)\operatorname{Var}(\ln\varphi_\omega)}{(\sigma-1)^2\operatorname{Var}(\ln\varphi_\omega)} \;=\; t\,.$$

regressive incidence $\Rightarrow \boldsymbol{\beta > 0}$ progressive $\Rightarrow \beta < 0$

proportional, selection $\Rightarrow \beta = 0$

From the theory to the data

- Nielsen scanner data, 2012–2019; balanced stores; event September 2016; reference year 2015; donut around straddle year 2016.
- **The firm:** brands rolled up to parent companies.
- **Baseline size, out of sample:** 2011 category revenue, the year before the panel.

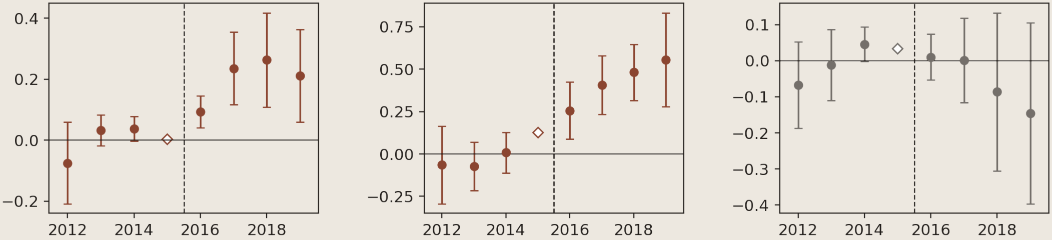
$$r_{ict} = \exp\left\{\beta\left(\ln \mathrm{size}_i^{2011} \times \mathrm{post}_t\right) + \delta_{ic} + \delta_{st}\right\} \cdot \eta_{ict} \qquad \text{(PPML; Wild Bootstrap SEs)}$$

β estimates the elasticity of incidence t. The state-by-quarter δ_{st} and the firm-county δ_{ic} effects absorb any general equilibrium effects.

The sample is products exposed to Salmonella, the pathogen behind the outbreak that justified FSMA: dry, low-moisture, ready-to-eat foods, long treated as low-risk because they are dry, and brought under mandatory preventive controls only after the Act.

Nuts

A nut **roaster** handles the exposed product (cooling and salting before packing): recontamination control. A handler of **raw** nuts offloads the duty to the supplier, and does not have to implement controls.

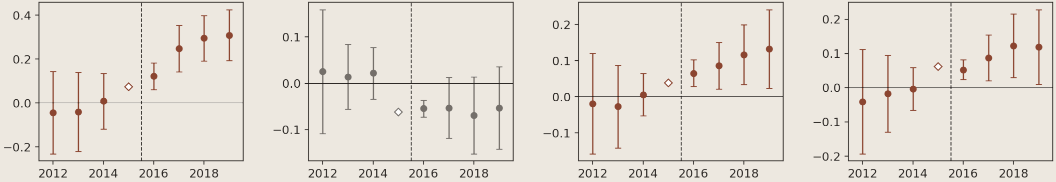


dry-roasted $\hat{\beta} = +0.238^{***}$ | oil-roasted $\hat{\beta} = +0.456^{***}$ | raw (placebo) $\hat{\beta} = -0.075$ (n.s.)

Oil-roasted: two firms selling ten-to-one in 2011 ended the post period selling nearly thirty-to-one.

Dried fruit and peanut butter

True dried fruit: contamination control runs through the whole line. **Fruit snacks**: terminal cook, packaged immediately after, nothing to monitor.
Peanut butter: the statute's origin category (no placebo axis).



true dried $\hat{\beta} = +0.292^{***}$

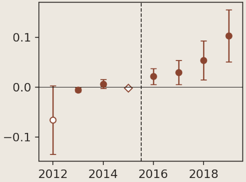
fruit snacks (placebo) $\hat{\beta} = -0.059$ (n.s.)

PB creamy $\hat{\beta} = +0.103^{***}$

PB chunky $\hat{\beta} = +0.097^{**}$

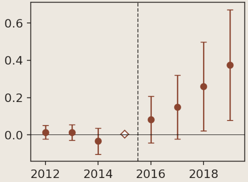
Linear pre-drift

Categories with a linear pre-drift; estimand = the trend break, the change in the gradient **per year**, net of the removed pre-drift.



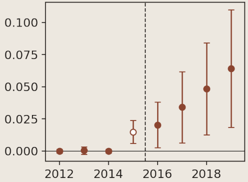
cereal

$+0.036^{***}$ /yr



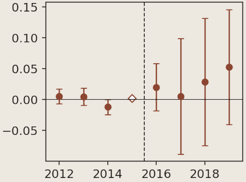
granola cereal

$+0.112^{***}$ /yr



seasoning

$+0.015^{***}$ /yr



flavored crackers

$+0.024^{**}$ /yr

The rest of the paper

- The mechanism in **general equilibrium**.
- The **welfare cost**, measured: Pareto marginal costs, where β maps into a change in the shape parameter.
- What the **raising-rivals'-costs** tradition would conclude about welfare reading these markets as homogeneous.

The mechanism in general equilibrium

Nested CES economy; labor the only factor; compliance is a labor cost. A single industry receives a regulation with regressive incidence. Under the standard assumption on substitution across industries, $\sigma_0 \leq 1$ (Atkeson & Burstein 2008, Hsieh & Klenow 2009; Comin, Lashkari & Mestieri 2021):

- **Expenditure flows toward the regulated industry**: its revenue and aggregate profit rise, paid by the other industries and by consumers.
- The most productive firms still win. They win more, and the winner set is weakly *larger* than in partial equilibrium.
- The regulation is a **welfare loss** for the economy.

A regressive incidence rule is collectively profitable for the regulated industry, individually profitable for the most productive, and a welfare loss for the economy.

Measuring the welfare cost

Parameterize marginal costs as Pareto with shape θ, written by rank ($\omega =$ the firm's quantile). With a constant elasticity of incidence, β estimates:

$$\psi_\omega \;=\; \psi_{\min}\,(1-\omega)^{-\frac{1}{\theta}} \qquad \Longrightarrow \qquad \psi^R_\omega \;=\; \big(1+\varepsilon_\omega\big)\,\psi_\omega \;=\; \psi_{\min}\,(1-\omega)^{-\frac{1+\beta}{\theta}}\,.$$

The estimated β is a shift of the shape parameter, $\theta \to \theta/(1+\beta)$: the cost tail fattens, and the industry concentrates.

- Calibration: β category by category; θ from the pre-period size distribution; demand elasticities from published estimates on the same Nielsen scanner data (Hottman, Redding & Weinstein 2016). The welfare cost then has a **closed form identified by** β.

Welfare using the current literature

The *raising rivals' costs* tradition works in a **homogeneous-good oligopoly**.

- The change in welfare is sign **ambiguous**: reallocation toward the low-cost leader, resulting from regressive incidence, is treated as an efficiency gain.
- The market share of the leader is a sufficient statistic for the differential in productivity against the followers.
- Thus, past a leader market share threshold, *rrc* reports the regulation as *welfare-improving*, because production went from the inefficient followers to the productive leader.

This threshold would be met in four of my categories:

	leader share	threshold
Nuts, dry-roasted (Wonderful)	64%	0.53
Crackers, flavored (Mondelez)	63%	0.54
Peanut butter, creamy (Jif)	57%	0.57
Seasoning (McCormick)	55%	0.53

A researcher studying FSMA using the homogeneous-good *rrc* tradition would conclude that it was welfare-improving in these four categories.

Takeaways

- A firm gains from a cost-raising rule exactly when the rule raises the price aggregator proportionally more than the firm's own cost.
- Regressive incidence is observable: a steepening of the productivity–size gradient. We measure the elasticity of incidence of FSMA in the low-moisture, ready-to-eat categories and pair them with placebos whenever available.
- Modeling these episodes with the homogeneous-good standard approach would report welfare gains where losses occurred.

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Why the within-industry size distribution matters

- Production has been concentrating in the largest businesses for at least a century: within industries, the size distribution has grown steadily more unequal. (Kwon–Ma–Zimmermann 2024)
- The fall of the labor share is not firms paying less — it is sales reallocating toward each industry's largest, low-labor-share firms. (Autor et al. 2020)
- Inequality between workers is, to a first approximation, inequality between firms: two thirds of the rise in earnings dispersion since the late 1970s. (Song et al. 2019)
- The aggregate markup rose not because the typical firm charges more (the median barely moved) but because share reallocated toward high-markup firms, mostly within industries; the welfare cost of that dispersion is first order. (De Loecker et al. 2020; Edmond et al. 2023)
- A fat-tailed distribution makes the economy less stable than the law of large numbers promises: shocks to the 100 largest firms account for roughly a third of output-growth fluctuations. (Gabaix 2011)

None of these literatures is about regulation. Each is a reason to care about the

Cost-function space: isolation and the reduction

Definition (isolated cost-minimization)

Firm i takes factor prices r as given and r is unaffected by the choices of any *individual* firm.

Proposition (reduction)

Under isolation, the firm's problem separates: compute $C_i(q_i)$ from own output and given factor prices; then maximize $p_i q_i - C_i(q_i)$. The industry equilibrium is characterized entirely by $\{C_i\}$ and demand.

- Consistent with r determined in equilibrium; what is ruled out is factor-market *power*.
- Constant returns + common overhead give the affine form $C_i(q) = c_i q + F$ with $c_i = e(r)/\varphi_i$ — a common *level* (neutral) and a cross-firm *pattern* (where redistribution lives).

Who wins: proof, and the boundary theorems

Proof of the theorem (one step): share and profit are strictly decreasing in ψ_ω/P (envelope); relative cost falls iff $\psi^R_\omega/\psi_\omega < \rho$.

Theorem (scale changes are neutral)

$\psi^R_\omega = r\psi_\omega$: by linear homogeneity $\rho = r$ — every firm exactly at the threshold; every share, revenue, and profit level unchanged. (Where the neutrality results of Macedoni–Weinberger and Anouliès live.)

Theorem (location shifts harm the most productive)

$\psi^R_\omega = \psi_\omega + a$: the ratio $1 + a/\psi_\omega$ is heaviest at low cost — there is a unique cutoff below which every top firm strictly loses.

Three sufficient forms, ordered by how much the top is spared

Corollary (kink: an untouched top)

$\psi^R_\omega = \psi_\omega$ on $(0,\bar{\omega})$, costs rise weakly elsewhere (strictly on positive measure) $\Rightarrow$ *every* firm in $(0,\bar{\omega})$ strictly gains — up to the kink point.

Corollary (vanishing burden)

$\rho > 1$ and $\lim_{\omega \downarrow 0} \psi^R_\omega / \psi_\omega = 1$ $\Rightarrow$ a neighborhood of the top strictly gains. (Pareto tail-fattening is the parametric case: the "shape-changing" name.)

Corollary (regressive incidence: a least-burdened top)

$\varepsilon_\omega \to \varepsilon_0$ at the top, $\varepsilon_\omega \geq \varepsilon_0$ a.e. (strict on positive measure) $\Rightarrow$ the top strictly gains. *No firm need be spared.*

Kink = the wage floor under rent-sharing: the compliance rule that codifies the top's

Exit amplifies

Corollary

Add an operating fixed cost $F > 0$ and suppose a positive measure of firms exits under the regulation. Then the winning firms that survive gain strictly *more* than in the no-exit case.

- With fewer active varieties, adding-up renormalizes shares over a smaller set — pushing the aggregator up *further*, the same direction that made the winners win.
- Exit is not a prerequisite of the mechanism — but when it occurs, it works for the winners, not against them.

CES–Pareto in closed form

CES demand (σ), Pareto costs $\psi_\omega = \psi_0(1-\omega)^{-1/\kappa}$:

$$\frac{\psi_\omega}{P} \;=\; \frac{(1-\omega)^{-1/\kappa}}{\mu\left[\kappa/(\kappa+\sigma-1)\right]^{\frac{1}{1-\sigma}}} \qquad \text{— the scale } \psi_0 \text{ cancels.}$$

A shape change is a marginal decrease in κ: every cost rises, the increase vanishes at the top. Firm ω benefits iff $\omega < \varepsilon$, with

$$\varepsilon \;=\; 1 - \exp\!\left(-\frac{\kappa}{\kappa+\sigma-1}\right), \qquad \varepsilon \downarrow \text{ in } \sigma, \quad \varepsilon \uparrow \text{ in } \kappa.$$

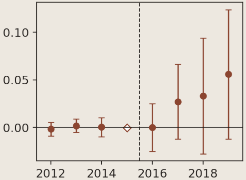
Design detail: selection criteria and threats

Five criteria, fixed before any post-period sign was examined: mechanism (dry RTE, *Salmonella* class, no terminal kill step); market coherence; gradient estimability; the partition rule (all buckets of a split axis enter — generates the treated/control contrasts); pre-trend validity (linear drift $\rightarrow$ trend-adjusted estimator, Freyaldenhoven–Hansen–Shapiro 2019). **Threats:**

- *Secular size trends* — estimator classes remove linear pre-drifts, leads displayed; within-category controls difference out the shared retail environment.
- *Anticipation* — final text published September 2015; reference year 2015.
- *Composition* — balanced stores under break-free retailers; 2016 crosswalk fixes firm boundaries; out-of-sample size + donut year keep treatment and baseline off the outcome path.
- *Cross-industry leakage* — nested CES appendix slide.

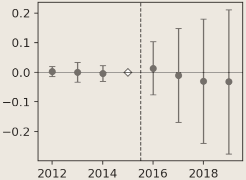
Chocolate and granola bars

Chocolate: an inclusion (nuts, PB) is an exposed RTE ingredient added *after* the kill step — the manufacturer carries supplier verification, in-plant handling, allergen controls. Plain chocolate's kill step sits upstream. **Granola bars**: same inherited burden with inclusions, plain oats-and-honey control; thin unit.



choc nut/PB

$+0.014^*$ /yr

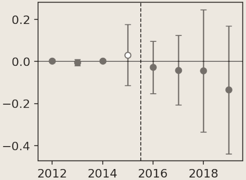


choc plain (control)

-0.011 (n.s.)

bars w/ incl.

$+0.035$ ($p = 0.39$)



bars plain (control)

-0.046 (n.s.)

Granola bars: control clean, treated directionally consistent, not significant on the estimand of record.

The politics sit exactly where the theory puts them

The industry's own association (Coca-Cola, PepsiCo, Kraft, General Mills, Nestlé) supported the bill, filed **40** lobbying disclosures, and celebrated the signing: *"the most comprehensive reform of our nation's food safety laws in more than 70 years."*

filer	category	filings
Blue Diamond Growers	nuts	50
American Peanut Shellers Association	peanut butter	41
General Mills / Kellogg / Post	cereal	16 / 12 / 2
American Bakers Association	flavored crackers	56

Opposition came from the other end of the size distribution (compliance would eat more than half of a very small operation's profits) — and won an exemption (Tester amendment).

GE rung 1: endogenous income, fixed mass

Shares are homogeneous of degree zero in Y: the who-wins characterization carries over *verbatim for shares*. For levels, under CES:

$$\Delta \ln r_\omega \;=\; (\sigma-1)\big[\Delta \ln P - \Delta \ln \psi_\omega\big] + \Delta \ln Y.$$

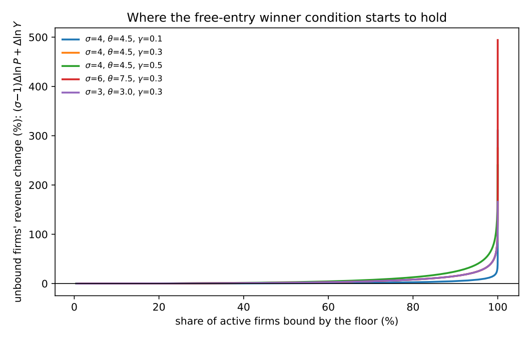
Proposition (regressive incidence in GE)

Cost share smallest at the top $\Rightarrow$ (i) top *shares* gain exactly, regardless of income; (ii) top *levels* change by $(\sigma-1)\big(\mathbb{E}_s[\varepsilon]-\varepsilon_0\big)+\Delta\ln Y$, first term strictly positive — gains whenever income does not contract by more than the relative-price gain.

First-order aggregator response $=$ the share-weighted average burden (CES, fixed mass). For one grocery category, Y barely moves.

GE rung 2: entry, and the calibration

With sunk entry (Melitz), adding-up renormalizes over an endogenous set: the variety margin can push the aggregator *down* for small deformations — the one rung where the guarantees lapse.



Calibration (rent-sharing free-entry economy, conventional grids): **the winner condition holds from the first binding floor in all 44 of 44 admissible cells.** At

Welfare: the matched comparison, closed form, and bounds

Homogeneous benchmark = a low-cost leader vs. n rivals, **Cournot**; discipline = a matched baseline (same price, leader share s_L, both margins — which pin σ and θ — and expenditure).

$$\Delta \ln W_{\text{diff}} = \frac{1}{\sigma - 1} \ln\Big[s_L + (1 - s_L)(1+t)^{1-\sigma}\Big] \in \Big[-(1 - s_L)\ln(1+t),\, 0\Big) < 0.$$

- Uniform markups $\Rightarrow$ reallocation welfare-neutral at the margin (Hulten in miniature); the loss is the direct resource cost.
- Cournot wedges are heterogeneous ($p - c_i = bq_i$): the allocative term = a *contraction* loss + a *reallocation covariance credit* toward the biggest-wedge firm. The gap between the two welfare numbers *is* this term.

Welfare: the dominance threshold and the understatement

Proposition (when the homogeneous benchmark reports a gain)

Homogeneous welfare rises with the marginal shape shock iff the leader's share exceeds

$$s_L^*(n) \;=\; \frac{n+4}{2(n+2)} \;\in\; \left(\tfrac{1}{2}, \tfrac{5}{6}\right].$$

- Elasticity-free; declining in the fringe; 0.53–0.57 at the application's firm counts — four categories cross.
- Below the flip, the small-shock understatement $U_0(s_L, n, \sigma)$ has a closed form; it rises to 100% at the threshold, and on calibrations disciplined by a common baseline never falls beneath a fifth.

Welfare: the second error — the survivor burden

Attribute an observed exit episode to a *fixed-cost* regulation when the truth is a shape regulation (both matched to the same exit cutoff). Variety losses cancel, and

$$\Delta W_{\text{shape}} - \Delta W_{\text{fixed}} \;=\; -\sum_{\text{survivors}} s_i\,\varepsilon_i \;<\; 0\,:$$

the **survivor burden**, which the fixed-cost accounting omits entirely — in its world, survivors' costs never moved. Sign-definite: matching on exit alone, the selection story *always* understates. The gradient detects exactly this error — the survivor burden is the object the fixed-cost story sets to zero.

Nested CES: cross-industry leakage

Outer elasticity σ_0 across industries, inner σ_g within; expenditure fixed (one food category is a negligible income share).

Proposition

Top firms' absolute profit rises iff $\sigma_g - 1 > (\sigma_0 - 1)(1 - S_g)$; sufficient: $\sigma_g > \sigma_0$ (the standard nesting assumption). The single-industry shock is the *worst* case.

Within-industry *relative* outcomes need no condition at all.

Microfoundations of the ordered burden

- **Factor intensity $\Rightarrow$ regressive burden.** Capital share $\alpha(\varphi)$ rising in productivity; a *uniform* labor tax has incidence $1 - \alpha(\varphi)$, declining in productivity. Increasing and *bounded* suffices — the vanishing burden is the corner case $\alpha \to 1$.
- **Rent-sharing wages $\Rightarrow$ kink** (Helpman–Itskhoki–Redding 2010). $w(\varphi) = w_0 \varphi^{\gamma}$: a wage floor binds only below a productivity cutoff — costs above it *literally unchanged*. (The Amazon case.)

The regulation is one instrument; the shape of its incidence is who uses the regulated factor.